



# REST & SOAP API Fundamentals

Week 5 • Day 3 • Technical Infrastructure & Modeling

TPM Academy



## Day 3 Objectives

- Use REST **resource modeling** — nouns, not verbs
- Apply HTTP **methods + status codes** consistently
- Design **idempotency** for safe retries
- Pick a **versioning strategy**
- Standardize **error semantics** with the 7 most-needed codes
- Know **when SOAP** is still the right call



# Today's Shape

| Clock         | Block                       | Outcome                              |
|---------------|-----------------------------|--------------------------------------|
| 09:00 – 09:15 | Opening                     | Pin TMD §§1–2; recap REST principles |
| 09:15 – 10:30 | Block 1 + Activity 1        | Resource modeling calibration        |
| 10:45 – 12:00 | Block 2 + Activity 2        | Design your feature's API            |
| 13:00 – 14:15 | Block 3 + Activity 3        | Idempotency, versioning, errors      |
| 14:30 – 16:00 | Block 4 + Activity 4 + Wrap | REST vs SOAP + AI critique           |

# Block 1 — REST & Resource Modeling

Nouns, not verbs



# REST Method Map

| Method | Purpose        | Idempotent?    |
|--------|----------------|----------------|
| GET    | Retrieve       | Yes            |
| POST   | Create         | No (typically) |
| PUT    | Replace        | Yes            |
| PATCH  | Partial update | No (typically) |
| DELETE | Remove         | Yes            |

**URLs are nouns** (resources). The verb is the HTTP method. POST /share is wrong; POST /shares is right.



# Status Codes That Carry Meaning

| Code        | Meaning         | When                                   |
|-------------|-----------------|----------------------------------------|
| 200         | OK              | Successful retrieval / update          |
| 201         | Created         | Successful creation                    |
| 204         | No content      | Successful with no body (e.g., DELETE) |
| 400         | Bad request     | Validation failed                      |
| 401         | Unauthenticated | Missing / invalid token                |
| 403         | Unauthorized    | Token valid; lacks permission          |
| 404         | Not found       | Resource doesn't exist                 |
| 409         | Conflict        | Idempotency / state collision          |
| 422         | Unprocessable   | Validated but business rule violated   |
| 429         | Rate limited    | Too many requests                      |
| 500/502/503 | Server side     | Different shapes of "us, not you"      |

# Activity 1 — Resource Calibration

Triads • 35 minutes

Design the resource model for a generic todo-list API (lists / items / collaborators).

- 10 min: sketch resources + URLs
- 10 min: methods per resource
- 10 min: trickiest design choice (sharing, reordering)
- 5 min: argue + readout prep



10 + 10 + 10 + 5

# Block 2 — Your Feature's API

From entities to endpoints





# Worked Endpoint (FieldPulse)

```
POST /v1/dispatchers/{dispatcher_id}/reconcile-events
```

<p>Headers:

Authorization: Bearer <SSO JWT, scope: reconcile.write>

Idempotency-Key: <client uuid=""></client></p>

<p>Body:

ticket\_ids: UUID[] (1..100)

submitted\_at: ISO 8601 timestamp

client\_metadata: { user\_agent, location }</p>

Responses:

201 Created – { id, ticket\_ids, submitted\_at }

400 Validation error – { error.code, fields[] }

401 Missing/invalid token

403 Token lacks reconcile.write scope

409 Idempotency-Key conflict

422 Tickets ineligible

429 Rate limited – Retry-After: <seconds></seconds>

## Activity 2 — Design Your API

Triads • 40 minutes

- 10 min: list resources from data model
- 10 min: design URL paths
- 10 min: methods + status codes per path
- 10 min: document one endpoint in detail



10 + 10 + 10 + 10

# Block 3 — Idempotency, Versioning, Errors

The aspects most often hand-waved



# Idempotency — Safe to Retry

## Natural

PUT and DELETE are idempotent by HTTP definition.

## Idempotency-Key header

Client sends a UUID; server stores key + response; returns same response on retry.

## Conditional

If-Match / If-None-Match gate the operation by version.

**For mutating endpoints:** decide approach + window (typical: 24 hours).



# Versioning Strategies

| Strategy            | Where version lives   | Pros                         | Cons                    |
|---------------------|-----------------------|------------------------------|-------------------------|
| URL path            | /v1/...               | Easy to read; clear breaking | Code duplication        |
| Header              | X-API-Version: 2      | URL stays clean              | Less discoverable       |
| Content negotiation | Accept: ...vnd...;v=2 | Standard                     | Often poorly understood |

**Default for B2B APIs: URL path versioning.**



## Error Body Shape

```
{
  "error": {
    "code": "ticket_not_eligible",
    "message": "Ticket cannot be reconciled in current state",
    "fields": [
      {"name": "ticket_ids[3]", "reason": "ticket already reconciled"}
    ]
  },
  "request_id": "req_01H...",
  "documentation_url": "https://docs.../errors/ticket_not_eligible"
}
```

- **code** — programmatic; clients switch on this
- **message** — human-readable
- **fields** — when validation failed
- **request\_id** — for support tickets

# Activity 3 — Idempotency, Versioning, Errors

Triads • 40 minutes

- 15 min: idempotency per mutating endpoint + window
- 10 min: pick versioning strategy
- 15 min: error body shape + 7 most-needed codes



15 + 10 + 15

# Block 4 — REST vs SOAP + AI Critique

When SOAP is still the right answer





## When SOAP Is Still the Right Call

- Integration partner only speaks SOAP (banks, government, EDI-adjacent)
- WS-Security or WS-ReliableMessaging contractually required
- Partner has invested heavily in SOAP tooling; rewrite is impractical

**For most new B2B features:** REST + JSON. SOAP appears occasionally for legacy enterprise integrations. Document the choice honestly.



# AI Critique Prompt

```
Role: API design reviewer with strong REST opinions.  
Context: <paste url="" paths="" methods="" status="" codes="" error="" body="" shape="" idempotency="" approach="" versioning="">  
Task: Identify the top 5 issues in this API contract.  
Constraints:  
- Be specific: cite the path, method, or convention  
- For each, suggest the correction  
- Avoid pedantic preferences (singular/plural is fine if consistent)  
Format: Numbered – Issue / What's wrong / Fix.</paste>
```

## Activity 4 — SOAP + AI Critique

Triads • 45 minutes • Wrap

- 10 min: SOAP question — any partner needs it?
- 10 min: AI critique → 5 issues
- 10 min: adopt / defer / reject
- 10 min: polish §3 + provenance



10 + 10 + 10 + 10 · 15-min wrap



# Day 3 Wrap

## What we built

- Resource design with URLs
- Methods + status codes
- One detailed endpoint
- Idempotency / versioning / error semantics
- SOAP exceptions documented
- TMD §3 drafted

## What opens tomorrow

- **High-level technical modeling**
- Sequence diagrams — happy + sad/weird
- Putting data + cloud + APIs together

**Bring to Day 4:** TMD §§1–3. Tomorrow we draw the sequences end-to-end.

# Questions & Reflection

Which of your endpoints would a hostile integration partner break first?